

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0617	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Y. Thanushma	Reg. No.:	192425355
Section / Batch:	slot A / Hostel / 2024	Date:	

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Questions	Marks
Application-Based Questions	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks

SECTION A – APPLICATION BASED QUESTIONS

- 1) A video compression algorithm follows the recurrence $T(n) = 8T(n/2) + n^3$. Use the Master Theorem to determine Time complexity. Identify parameters and justify the case. Discuss how processing Time increases with video resolutions.
- 2) Application : Search Engine Indexing
- 3) Application : AI Based Chatbot processing

Code of Conduct 192425355
 I, V. Tharushma (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
 Signature of the Student: V. Tharushma

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding: some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Y. Thanushma
Number :- 192425355
Code :- CSA0617

Course Name :- Design
And Analysis of
Algorithms
Course Faculty :-
Dr. S. Pradeep Kumar

CO-1 Assessment 1

Application :- Video Compression Algorithm

A video compression algorithm follows the recurrence: $T(n) = 8T(n/2) + n^3$. Use the Master Theorem to determine Time Complexity. Identify parameters and justify the case. Discuss how processing time increases with video resolutions.

Problem Context :-

A video compression algorithm recursively divides a video into a smaller parts for compression. The recurrence relation is:

$$T(n) = 8T(n/2) + n^3$$

We use the Master Theorem to determine its time complexity and analyses how processing time changes as video resolution increases.

Applying the Master Theorem:-

Step 1 :- Identify the parameters

$$a = 8$$

$$b = 2$$

$$f(n) = n^3$$

Step 2:- Calculate

$$n^{\log_b a} = n^{\log_2 8} = n^3$$

Step 3:- Compare

$$f(n) = \Theta(n^3) = \Theta(n^{\log_b a})$$

Since $f(n)$ equals $n^{\log_b a}$, Master Theorem case 2 applies

Final Time Complexity :-

$$T(n) = \Theta(n^3 \log n)$$

Justification :-

The work performed outside the recursive calls grows at the same rate as the recursive work. Therefore, the extra $\log n$ factor appears resulting in $\Theta(n^3 \log n)$

Practical Analysis:-

As video resolution increases, the number of pixels also increases. Since the algorithm runs in $\Theta(n^3 \log n)$ time, processing high resolutions videos (such as 4k and 8k) requires sustainability and more computation, making compression slower.

Application :- Search Engine Indexing

A search engine indexing algorithm is defined as $T(n) = 2T(n/2) + n^2$. Apply the Master Theorem and determine the Time Complexity. Clearly explain the chosen case. Analyze how indexing time grows with web pages.

Problem Context:

A search engine builds an index for web pages using the recurrence:

$$T(n) = 2T(n/2) + n^2$$

The objective is to determine the algorithm's time complexity using the Master Theorem and explain how indexing time grows.

Applying the Master Theorem:-

Step 1: Identify the parameters

$$a = 2$$

$$b = 2$$

$$f(n) = n^2$$

Step 2:- Calculate

$$n \log_b a = n \log_2 2 = n$$

Step 3:- compare

Since

$$n^2 = \Omega(n^{1+\epsilon}) \quad \text{where } \epsilon = 1, \text{ and the regularity condition is satisfied,}$$

$$\boxed{2(n/2)^2 = \frac{n^2}{2} \leq cn^2, \quad c < 1}$$

Master Theorem Case 3 Applies

Final Team Complexity,

$$T(n) = \Theta(n^2)$$

Justification

The non recursive work n^2 dominates the recursive work, so the total running time is determined by $f(n)$

Practical Analysis:-

As the number of web pages increases, indexing time increases quadratically. Therefore, search engines handling millions of web pages require significant computational resources and optimization techniques.

3) Application : AI - Based Chatbot processing
An AI chatbot processes user queries using:
 $T(n) = T(n/2) + \log n$. Apply the Master Theorem and determine the Time complexity.

Problem Context :-

An AI chatbot processes user requires according to recurrence:

$$T(n) = T(n/2) + \log n$$

The goal is to determine the time complexity using the Master Theorem and explain its effect on chatbot response time.

Applying the Master Theorem:-

Step 1 :- Identify the parameters

$$a = 1$$

$$b = 2$$

$$f(n) = \log n$$

Step 2 :- Calculate

$$n^{\log_b a} = n^{\log_2 1} = 1$$

Step 3 :- Compare

Since

$$f(n) = \Theta(n^0 \log^1 n)$$

this satisfies Master Theorem Case 2 (where $k=1$)

Final Time Complexity

$$T(n) = \Theta(\log^2 n)$$

Justification :-

The recursive work is constant, while the non-recursive work is algorithmic. According to Case 2 of the Master Theorem, the total complexity becomes $\Theta(\log^2 n)$.

Practical Analysis:-

As the number of user requires queries on input size increases, the chatbot's response time grows very slowly ($\log^2 n$). This enables AI chatbots to maintain fast and efficient responses even when processing large amounts of data.